When Can a Higher Loan Spread Predict Lower Delinquency?

Endogenous screening under borrower opacity · Track B

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github.com/fgalarzac/ai-project

When does a bank's relative loan spread predict a safer rather than a riskier future portfolio?

Object observed today | **Outcome observed later**

$$D_{jst} = i_{jst} - \text{median}_{k \neq j}\, i_{kst} \qquad m_{js,t+h} = \text{delinquency rate}$$

Track B: formalize the missing bank problem behind my thesis on Peruvian commercial banking.

1 · Why the sign is not obvious

selection channel

A higher relative price loses comparison shoppers and can leave a worse applicant pool.

$$\frac{\partial q^0}{\partial D} = a(\kappa) > 0$$

screening channel

A higher performing-loan payoff raises the return to screening and can improve booked quality.

$$\frac{\partial e^*}{\partial D} > 0$$

Borrower opacity changes which mechanism dominates.

2 · Model: timing and primitives

1. Segment opacity $\kappa \in [0, 1]$ is observed.
2. The bank posts $D \in [\underline{D}, \overline{D}]$ and faces demand $L(D, \kappa)$, with $L_D < 0$.
3. Its applicant pool has pre-screen default probability

$$q^0(D, \kappa) = \bar{q} + a(\kappa)D.$$

4. The bank chooses screening $e \in [0, \bar{e}]$, producing

$$q(D, e, \kappa) = q^0(D, \kappa) - g(\kappa)e.$$

A performing loan pays $R(D) = \bar{R} + D$. Screening costs $c(\kappa)e^2/2$ per loan.

2 · Model: the bank's problem

$$\max_{D\in\mathcal{D}} L(D,\kappa)\left\{\max_{e\in[0,\bar{e}]} \underbrace{[1-q^0(D,\kappa)+g(\kappa)e](\bar{R}+D)}_{\text{expected repayment}} - F - \frac{c(\kappa)}{2}e^2\right\}.$$

- The compact price set closes the first stage.
- The proposition concerns $q^*(D,\kappa)$, the quality map conditional on an observed price.
- This is a forecasting statement, not a causal effect of forcing a bank to change D.

3 · Screening FOC and closed form

$$\frac{\partial \pi}{\partial e} = g(\kappa)(\bar{R} + D) - c(\kappa)e = 0, \qquad \frac{\partial^2 \pi}{\partial e^2} = -c(\kappa) < 0.$$

$$\boxed{e^*(D, \kappa) = \frac{g(\kappa)}{c(\kappa)}(\bar{R} + D)}$$

$$\boxed{q^*(D, \kappa) = \bar{q} + a(\kappa)D - \underbrace{\frac{g(\kappa)^2}{c(\kappa)}}_{h(\kappa)}(\bar{R} + D)}$$

$h(\kappa)$ is screening leverage, not screening effort itself.

3 · Candidate Proposition 1

Assume interiority, feasible probabilities, continuity, and

$$a'(\kappa) < 0, \qquad h'(\kappa) > 0, \qquad a(0) > h(0), \quad a(1) < h(1).$$

Then a unique $\kappa^* \in (0,1)$ solves $a(\kappa^*) = h(\kappa^*)$, and

$$\boxed{\beta(\kappa) \equiv \frac{\partial q^*}{\partial D} = a(\kappa) - h(\kappa)}$$

is positive below κ^*, zero at κ^*, and negative above it. Moreover,

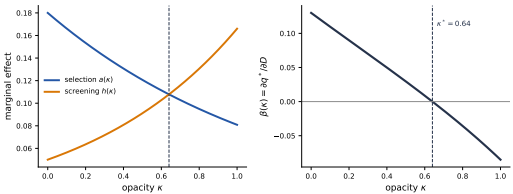
$$\frac{\partial^2 q^*}{\partial D \partial \kappa} = a'(\kappa) - h'(\kappa) < 0.$$

3 · What the key condition really says

$$h(\kappa) = \frac{g(\kappa)^2}{c(\kappa)}, \qquad h' > 0$$

is equivalent to

$$2\frac{g'}{g} > \frac{c'}{c}.$$



Correction to the thesis narrative

Higher screening cost alone cannot create the negative slope. Opacity must raise the marginal information gain enough to dominate its cost.

3 · From origination quality to future delinquency

Let future observed delinquency satisfy

$$m_{j,s,t+h} = \rho_h q^*_{j,s,t} + \varepsilon_{j,s,t+h}, \qquad \rho_h > 0.$$

Then

$$\text{sign}\left(\frac{\partial \mathbb{E}_t[m_{j,s,t+h}]}{\partial D_{j,s,t}}\right) = \text{sign}[a(\kappa_s) - h(\kappa_s)].$$

- D is computed within product, maturity, currency, and month.
- κ_s is an ordinal segment-opacity proxy, not a directly estimated structural primitive.
- The empirical exercise compares out-of-sample forecasts.

4 · Why this is not already solved

Varian (1980)
Shoppers and captive buyers generate price dispersion.
Gap: no screening or default.

Broecker (1990)
Imperfect tests interact with rate competition.
Gap: testing is not a continuous effort choice.

Ruckes (2004)
Screening responds to business-cycle borrower quality.
Gap: no opacity threshold for the relative-price signal.

Dell'Ariccia and Marquez (2006)
Interbank information asymmetry selects screening or pooling.
Gap: no continuous $\partial q^*/\partial D$ decomposition.

Checked: model sections and appendices, the working-paper record, and the IMF reprint.

5 · Proof, simulation, and Lean plan

1. **Proof:** Proposition 1, then a robustness result for general increasing $R(D)$.
2. **Simulation:** map the parameter region where $a - h$ crosses zero and verify interiority and probability bounds.
3. **Forecasting:** evaluate whether D improves predictions of m_{t+h} by segment opacity and horizon.
4. **Lean:** formalize the threshold theorem after its assumptions and statement are frozen.

Already checked in code: FOC, SOC, monotonicity, unique crossing, effort bounds, and $q^* \in (0, 1)$.

5 · Main risk and disciplined fallback

Most likely failure

The maintained condition $h'(\kappa) > 0$ may fail in a richer signal technology or in calibration.

Fallback result

Do not force a reversal. Characterize the three identified regions

$$a(\kappa) > h(\kappa), \qquad a(\kappa) = h(\kappa), \qquad a(\kappa) < h(\kappa),$$

and treat threshold existence as an estimable restriction.

The contribution is the condition for the sign, not a guaranteed negative coefficient.